Q.27 Explain the reversible and irreversible reaction along with the suitable example for each type?

Q.28 Discuss the relationship between concentration and conversion for constant volume batch reactor?

Q.29 What is difference between rate of reaction and reaction rate constant?

Q.30 Discuss the temperature dependency of rate equation using collision theory?

Q.31 Describe the catalyst accelerators in brief?

Q.32 Explain the regeneration of catalyst with the help of suitable example?

Q.33 Explain the concept of activation energy in brief?

Q.34 Describe the differential method for analysis of rate data in brief?

Q.35 Discuss the temperature dependency of rate equation using collision theory?

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Describe principle and working of fluidized bed reactor with the help of suitable diagram in detail?

Q.37 Define zero order reaction? Explain the determination of order and rate constant for zero order reaction using integral method with the help of suitable diagram?

Q.38 Explain the effect concentration, pressure and temperature on the state of chemical equilibrium using Le-chatlier principle, in detail?



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Roll No. .................

**5th Sem / Chem, Chem Engg (Spl. Paint Tech.)**
**Chem Engg. (Spl. Polymer Engg.)**

**Subject:- Chemical Reaction Engineering**

Time : 3Hrs. M.M. : 100

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The rate of a reaction increases, if concentration of reactants is?

a) increased b) decreased

c) always unaffected d) uncertain

Q.2 Half life period for first order reaction is related to initial concentration of the reactant as?

a) directly proportional

b) inversely proportional

c) independent

d) none of these

Q.3 The rate constant of a reaction depends upon?

a) mass b) time

c) weight d) temperature

Q.4 A reaction which proceeds with evolution of heat is called?

a) endothermic reaction

b) exothermic reaction

c) elimination reaction

d) substitution reaction

Q.5 Integral method for analyzing the kinetic data is used?

a) when data are scattered
b) for testing specific mechanism
c) both (a) and (b)
d) none of these

Q.6 The equation in which rate does equation does not corresponds to a stolchiometric equation is called?
a) elementary reaction
b) non-elementary reaction
c) catalytic reaction
d) non-catalytic reaction

Q.7 A reaction in which one of the products of reaction acts as catalyst is called?
a) homogeneous catalytic reaction
b) heterogeneous catalytic reaction
c) autocatalytic reaction
d) biochemical reaction

Q.8 With the increase in temperature, the rate constant obeying Arhenious law?
a) increases b) decreases
c) remains constant d) none of these

Q.9 The material which improves the activity of a catalyst is called?
a) carrier b) inhibitor
c) modifier d) promoter

Q.10 Plug flow reactor is characterized as flowing systems of?
a) spherical geometry b) cylindrical geometry
c) plain geometry d) none of these

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What are inhibitors?



Q.12 Name the type of reaction if it takes place in one phase alone?

Q.13 Which state is used to describe the condition where rate of forward reaction becomes equal to rate of backward reaction?

Q.14 Define space velocity?

Q.15 Write full form of CSTR?

Q.16 Write any one type of steady state flow reactor?

Q.17 Define the catalyst?

Q.18 What are accelerator in catalysis?

Q.19 Half life period is independent of initial concentration of reactant in case of which type of order of reaction?

Q.20 Which term is used to describe a substance whose addition to the reaction mixture increases the rate of chemical reaction?

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Discuss the plug flow reactors in brief with the help of neat diagrams?

Q.22 What is difference between homogeneous reaction and heterogeneous reaction?

Q.23 State Le-chatlier principle? Discuss its significance in brief?

Q.24 Define and discuss the molecularity of reaction in brief?

Q.25 Define the half life period of a reaction? Discuss its importance in brief?

Q.26 What is difference between catalyst promoter and catalyst poison?